BAS-151- Physics Lab

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List of Experiments

Any ten experiments, at least four from each group.

Group –A

1. To determine the wavelength of monochromatic light by Newton‟sring.
2. To determine the wavelength of monochromatic light with the help of Fresnel‟s biprism.
3. To determine the focal length of two lenses by nodal slide and locate the position of cardinal points.
4. To determine the specific rotation of cane sugar solution using polarimeter.
5. To determine the wavelength of spectral lines using plane transmission grating.
6. To study the polarization of light by simple reflection usinglaser.
7. Measurement of Wavelength of a laser (He- Ne) light using single slit diffraction.

Group – B

8. To determine the specific resistance of a given wire using Carey Foster‟sbridge.
9. To study the variation of magnetic field along the axis of current carrying - Circular coil and then to estimate the radius of thecoil.
10. To verify Stefan‟s Law by electrical method.
11. To calibrate the given ammeter and voltmeter by potentiometer.
12. To study the Hall effect and determine Hall coefficient, carrier density and - mobility of a given semiconductor using Hall effect setup.
13. To determine the energy band gap of a given semiconductor material.
14 To determine E.C.E. of copper using Tangent or Helmholtz galvanometer.
15. To draw hysteresis curve of a given sample of ferromagnetic material and from - this to determine magnetic susceptibility and permeability of the given specimen.
16. To determine the ballistic constant of a ballistic galvanometer.
17. To determine the coefficient of viscosity of aliquid.
18. Measurement of fiber attenuation and aperture offiber.
19. High resistance by leakage method.
20. Magnetic Susceptibility of paramagnetic solution.